

ADIL QURESHI

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Experience

RoadGauge Ltd.

Lahore, Pakistan

Computer Vision Engineer

Aug 2022 – July 2024

- Developed an low-cost image-based road surface inspection product from the ground-up. Achieved surface defect measurement **accuracy of sub 4mm** from a monocular road facing video.
- Trained a custom DeepLabv3+ for lane **detection** and **segmentation** of pavement defects. Achieved **iOU** of **91.34%**.
- Developed Structure from Motion (**SfM**) pipeline for **3D reconstruction** of road surface and camera **pose estimation** involving SIFT key-point matching, 8-Point algorithm for **fundamental matrix** estimation and linear triangulation.
- Developed algorithms for perception and **depth estimation** for generating **RGB-D images** and stitched **2D birds-eye-view** images of road surface, involved **point-cloud transformations**.
- Implemented RANSAC planar surface segmentation algorithm for **point-clouds** to measure depth-based defects.
- Developed algorithms for **IMU and GPS** data processing for robust point-cloud scale estimation.

RoadGauge Ltd.

Lahore, Pakistan

Computer Vision Intern

May 2022 – Aug 2022

- Improved existing algorithm for **object tracking** and inventory of elements of interest such as traffic lights and signs in dynamic cluttered environment **by 11%** using classical **computer vision** and deep learning approaches.

Deep Learning Lab, NCAI

Islamabad, Pakistan

Research Assistant

May 2021 – Aug 2022

- Worked on **classification** and localization of abnormalities in EEG signals for assisting neurologists in diagnosis. Leveraged **predictive modeling**, **statistical analysis** and **exploratory data analysis**.
- Developed CWT (continuous wavelet transform) based **pre-processing routine** to generate scalograms for training **convolution neural networks (CNN)**. Led to **25% increase** in performance compared to initial baseline.
- Trained and evaluated classical and deep learning based **classification architectures** (GoogLeNet, VGG16, EfficientNetB0, Random Forest Classifiers, Decision Trees, SVM).
- Curated and **published** the first-of-its-kind event localization EEG dataset with a focus on South Asian demographics.

Education

University of Maryland

College Park, MD

Master of Engineering in Robotics

Sep 2024 – May 2026 (expected)

- Coursework: Perception for Autonomous Robots, Planning for Autonomous Robots, Python Applications for Robots*

National University of Sciences and Technology (NUST)

Islamabad, Pakistan

Bachelor of Engineering in Electrical Engineering

Sep 2018 – Aug 2022

Skills

Programming: Python, C, C++, MATLAB

Frameworks and Libraries: PyTorch, TensorFlow, OpenCV, Open3D, PCL, ROS2, Numpy, Scikit-Learn, Matplotlib

Tools: Linux (Terminal Commands, Bash/Shell, Git), JIRA, Docker (DevOps), CI/CD, VS Code, Jupyter Notebook

Projects

Neural Radiance Fields (NeRF) for 3D Scene Rendering | *Python, PyTorch, Numpy, CUDA*

- Implemented and optimized the original NeRF paper for high-quality 3D scene reconstruction.
- Implemented dynamic frequency selection for positional encoding, adaptive sampling based on scene complexity.

Camera pose based 3D Object Tracking and Inventory | *Python, OpenCV, Open3D, Numpy*

- Used segmentation mask to identify 2D ROI and extracting corresponding 3D points from point-cloud. Applied voxel downsampling for optimization of large 3D reconstructions.
- Leveraged SfM-based camera pose information and frame correspondence for 3D object inventory management.

Speech to Image Generation Using Fine-tuned Latent Diffusion Model | *Python, Pytorch, CUDA*

- Fine-tuned Stable Diffusion v2 on custom dataset of subject to generate personalized images based on the prompt given.
- Employed OpenAI Whisper model API for speech-to-text generation.

Anomaly Detection in Streaming Energy Meter Time Series Data | *Python, PyTorch, TensorRT*

- Developed LSTM autoencoder pipeline for unsupervised time-series anomaly detection
- Acquired and pre-processed real-time input data stream from a energy monitor on Nvidia Jetson Nano.

Publications

- T.Y. Azim, M. Rahman, A. Abed, **A. Qureshi**, A. Vakeel, M. Ashraf, A. Saeed, A. Saeed, Z. Ul Mustafa "Low-cost automated image based rutting identification and measurement." *8th ICBMP (2024)*, doi.org/10.1201/9781003402541
- Alqarni, M.A, **Qureshi, A.J.**, and Alvi, M. "NeuroAssist: An Open-Source Tool for Automatic Event Detection in Scalp EEG", IEEE Access, [doi:10.1109/ACCESS.2024.3492673](https://doi.org/10.1109/ACCESS.2024.3492673)